Some notes for what I hope will be a good first class day

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Contents

1 Very basic introductions

2 First class task/ice-breaker

2.1 Divide class into teams of four or five each.

2.2 Assign to each team a label:

- cognitive psychologist
- experimental neuroscientist
- computational neuroscientist
- philosopher of mind

2.3 Consider the following question from the point of view of your assigned role:

An X learns that a cue Y predicts a reward Z. What evidence would you want to see to support this claim? What would constitute an ”explanation” for such a relation?

2.4 Give time for developing an answer.

2.5 Review the suggestions.

2.6 Now consider the following and tell me what changes:

- X = Dog; Y = Bell; Z = food.
- X = Pea Plant ; Y = air puff ; Z = light link
- X = Neuron ; Y = another neuron’s action potential; Z = third neuron’s action potential

3 Rescorla Wagner Equation

$$\Delta V_X^{n+1} = \alpha_X \beta (\lambda - V_{\mathrm{tot}})$$

and

$$V_X^{n+1} = V_X^n + \Delta V_X^{n+1}$$

where

- ΔV_X is the change in the strength, on a single trial, of the association between the CS labelled "X" and the US
- α is the salience of X (bounded by 0 and 1)
- β is the rate parameter for the US (bounded by 0 and 1), sometimes called its association value
- λ is the maximum conditioning possible for the US
- V_X is the current associative strength of X
- V_{tot} is the total associative strength of all stimuli present, that is, X plus any others

3.1 Is this an explanation? Is this a theory? What do we learn by changing our perspective to include equations?

3.2 What are Marr's Levels?

- Computational
- Algorithmic
- Implementational

4 LLMs

4.1 Are the a good model system for the mind? That is, does an LLM think like you do?

4.2 Are they a good tool for studying the mind?

4.3 How do we best use LLMs as a tool for our learning? How should we change our classroom experience?

4.4 First Assignment

4.4.1 Two sets of groups

Half of people in teams of 4. Half the people solo. Teams of four, please don't use LLMs or even an internet search. Library books are okay. Just talk it through with your group and each of you submit an answer. The other ”solo” half. Please don’t discuss your answer or ideas with anyone other than an llm. Answer the same questions, but in addition you will need to provide info on which LLM you used and the prompts (or even the entire chat if you want).

We want to help decide ourselves how to make the most constructive use of these tools.

Bring pencil/pen and paper to the next class. There will be some writing that you have to turn in.

5 Second Assignment Next Day of Class

Also in class. Show them the slime mold slides. It seems to fit our case of an X learning Y predicts Z, but it does it with no brain and an external memory. Is this a mind? Are its activities *cognitive*?

Write a 1/2 to 1 page essay. Does a slime mold think?

6 Third Activity – third class day

Discuss some of the essays and what we can see when comparing AI aided prep to conversation guided prep.